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A Study on User Experience of Automotive HUD Systems: Contexts of Information Use and User-Perceived Design Improvement Points

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ABSTRACT

Despite three decades of use and related research efforts, how automotive HUD can be designed to best serve the driver is still an on-going research issue. For any human-machine system, studying the experiences of the existing users is an important first step towards design improvement and innovation. Nonetheless, surprisingly few studies have investigated the user experience of the existing commercial automotive HUD systems. In an effort to address the research gap, the current study conducted a user survey on the usage contexts and design improvement points of the existing automotive HUD systems focusing on eleven high-priority HUD information items selected among those displayed by the existing commercial HUD systems. A total of fifty-one drivers with significant prior HUD use experience participated. Design implications were derived from the survey responses. They may be useful for improving the existing automotive HUD systems in both the system functions and the interface/interaction design.

1. Introduction

As a component of in-vehicle information system, automotive head-up display (HUD) system presents various information items, such as current speed, speed limit, navigation cues, safety warnings, etc., directly on the driver's forward field of view (FFOV). By doing so, it allows the driver to obtain the required information while keeping their eyes forward, instead of lowering their gazes away from the road scene ahead as in using conventional head-down displays (HDD) (Doshi, Cheng, & Trivedi, 2009; ISO/TR 16352, 2005; Liu, 2003; Liu & Wen, 2004; Stevens, Quimby, Board, Kersloot, & Burns, 2002; Ward & Parkes, 1994; Yang et al., 2018). Well-designed automotive HUD is believed to provide advantages over HDD, including less gaze and focal transition between the road scene and displays, and, possibly, faster detection of displayed information and outside events (Gish & Staplin, 1995; Liu, 2003; Liu & Wen, 2004; Oh, Ko, & Ji, 2016; Sojourner & Antin, 1990).

After three decades of automotive HUD use and related research efforts, the question of how to design the automotive HUD system to best serve the driver still remains an on-going research challenge. As in other man-made systems, the current automotive HUD systems could be further improved in terms of the system functions and the interface/interaction design so as to offer better utility and usability.

For any existing human-machine system, examining the current use practices and the problems and issues perceived by the users is an essential step towards design improvement and innovation; its importance has been repeatedly

emphasized in the product design and user experience literature (Hartson & Pyla, 2012; ISO 9241-210, 2010; Yayici, 2014). The design of automotive displays, including HUD, would be no exception to this. According to Bhise (2011), studying the actual usage conditions and driving situations when automotive displays are used can provide insights into the driver's tasks, informational needs, time constraints, and environmental conditions.

Despite the importance, however, little knowledge seems currently available concerning the context of system use and user-perceived problems/challenges for the existing automotive HUD systems. Numerous human factors (HF) and HCI studies have been conducted on automotive HUD systems – the existing HF/HCI studies on automotive HUD may be classified largely into four groups in terms of their purposes: 1) demonstrating the advantages of HUD over conventional HDD (Liu, 2003; Liu & Wen, 2004; Oh et al., 2016; Rutley, 1975; Sojourner & Antin, 1990; Srinivasan & Jovanis, 1997), p. 2) developing novel automotive HUD functions/applications (Bark, Tran, Fujimura, & Ng-Thow-Hing, 2014; Charissis & Papanastasiou, 2010; Doshi et al., 2009; Fujimura, Xu, Tran, Bhandari, & Ng-Thow-Hing, 2013; Haeuslschmid, Schnurr, Wagner, & Butz, 2015; Kim, Wu, Gabbard, & Polys, 2013; Park, Park, Won, Kim, & Jung, 2013; Tran, Bark, & Ng-Thow-Hing, 2013; Yoon, Kim, Baek, & Park, 2014), p. 3) comparing HUD interface design alternatives (Gregoriades & Sutcliffe, 2018; Guo, Zhao, Wang, & Jiang, 2014; Huang, Chao, Tsai, & Hung, 2013; Pfannmueller, Kramer, Senner, & Bengler, 2015; Plavšić, Duschl, Tönnis,

Bubb, & Klinker, 2009; Watanabe, Yoo, Tsimhoni, & Green, 1999) and, p. 4) determining the relative importance of different HUD information items (Beck & Park, 2018; Bergman, 2011; Guo et al., 2014; Huang et al., 2013; Moon & Park, 1998; Park, Kim, & Chong, 2012). However, the existing studies rarely dealt with the system use practices and the problems/issues perceived by the users.

Addressing the current knowledge gap would greatly benefit the design of future automotive HUD systems. Therefore, the objective of the current study was to examine the system use contexts (HUD information use contexts) and the user-perceived design improvement points for various information items displayed by the existing commercial automotive HUD systems. Eleven high-priority HUD information items were considered, which were selected among those displayed by the existing commercial HUD systems. A survey was conducted with experienced automotive HUD users to accomplish the research objective. User surveys have been used to understand product use practices and user wants and needs in a variety of product design contexts (Goodman, Kuniavsky, & Moed, 2012; Rubin & Chisnell, 2008). In light of the research objective, recruiting participants with significant actual automotive HUD use experience was deemed essential. Bergman (2011) showed that drivers' attitudes and perceptions towards the HUD technology could significantly change before and after actual HUD experience and further recommended that user studies on automotive HUDs must recruit subjects with extensive HUD experiences who feel HUD as a natural part of their driving experience.

2. Method

A total of fifty-one drivers participated in this study. Their demographic data are summarized in Table 1. As shown in Table 1, the participants had sufficient prior experience of using commercial automotive HUD products (average experience: 2.59 years).

Prior to this study, the study participants had subjectively evaluated the importance of twenty-two information items displayed by the existing commercial HUD systems using a five-point semantic differential scale (1-“Absolutely unimportant”, 2-“Unimportant”, 3-“Neutral”, 4-“Important”, 5-“Absolutely important”). Eleven information items had average importance ratings greater than three (“Neutral”). They (and their average importance ratings scores) were: current speed (4.73), speed limit (4.59), turn-by-turn navigation (4.37), maintenance warning (3.98), cruise control status (3.80), audio player status (3.45), traffic sign (3.35), distance to destination (3.25), gear position (3.24), RPM (3.24) and low fuel warning (3.02).

All of the eleven information items, except for audio player status, pertain to the primary driving tasks defined by Bubb

(2011). Bubb (2011) defined primary driving tasks as all the tasks necessary for drivers to keep the vehicle on the road and to proceed to the planned driving course (e.g. wayfinding, steering, accelerating, braking, shifting gears, and managing distance to other cars). Tonnis, Broy, and Klinker (2006) reported that the information items needed for the primary driving tasks would be best shown by HUD in the windshield area among different display zones.

The eleven information items also corresponded to the top half of the twenty-two items in terms of the number of participants with ‘prior information item experience’ – it ranged from 12 to 51, with an average of 28.4 for the top eleven information items, and from 2 to 12, with an average of 6.10 for the others. Note that although all of the fifty-one participants had at least one year of experience in using HUD system(s), not every participant had experienced all the twenty-two information items, since the existing HUD systems display different subsets of them. Participant's prior information item experience was crucial in that the research objective was to examine contexts of use and user-perceived design improvement points for various HUD information items.

In light of the above, the eleven high-priority HUD information items in terms of the perceived importance, relevance to primary driving tasks and number of participants with prior information item experience, were considered in the current study for examining the context of use and design improvement points.

For each of the eleven HUD information items, the participants who had prior experiences of using HUD systems displaying the information item were instructed to respond to two survey items. The two survey items were: (1) Describe the context of information use, and (2) Describe possible design improvement points (of the HUD system[s] that you have used).

For each HUD information item, the survey responses (comments) of the participants were examined and were refined into simple statements, each of which contained one simple concept, idea or fact. For each statement, the associated frequency (the number of participants who made the statement) was recorded. An affinity diagram analysis was conducted on the simple statements dataset to identify themes or topics (Figure 1). A total of three vehicle ergonomics experts with a combined research experience of twenty years participated in the affinity diagram analysis, which was conducted based on the process below:

- (1) Examining each statement and discussing its meaning.
- (2) Exploring for the common characteristics/similarities across the statements.
- (3) Grouping the statements that have an affinity/topical similarity into a cluster.
- (4) Making a label for each cluster to help assign new statements appropriately.
- (5) Modifying (creating/growing/merging/splitting) the clusters until all the statements are assigned properly.
- (6) Finalizing the labels and identifying themes or topics.

The procedure for generating the simple statements dataset and the affinity diagram analysis process described above were

Table 1. Demographic data of the 51 participants.

	Age (years)	Driving experience (years)	HUD use experience (years)	Mean driving time per day (hours/day)
Average	36.0	13.7	2.59	1.97
Range	23 ~ 50	3.5 ~ 25	1 ~ 10	0.5 ~ 5
STDEV	6.82	6.25	2.00	0.98

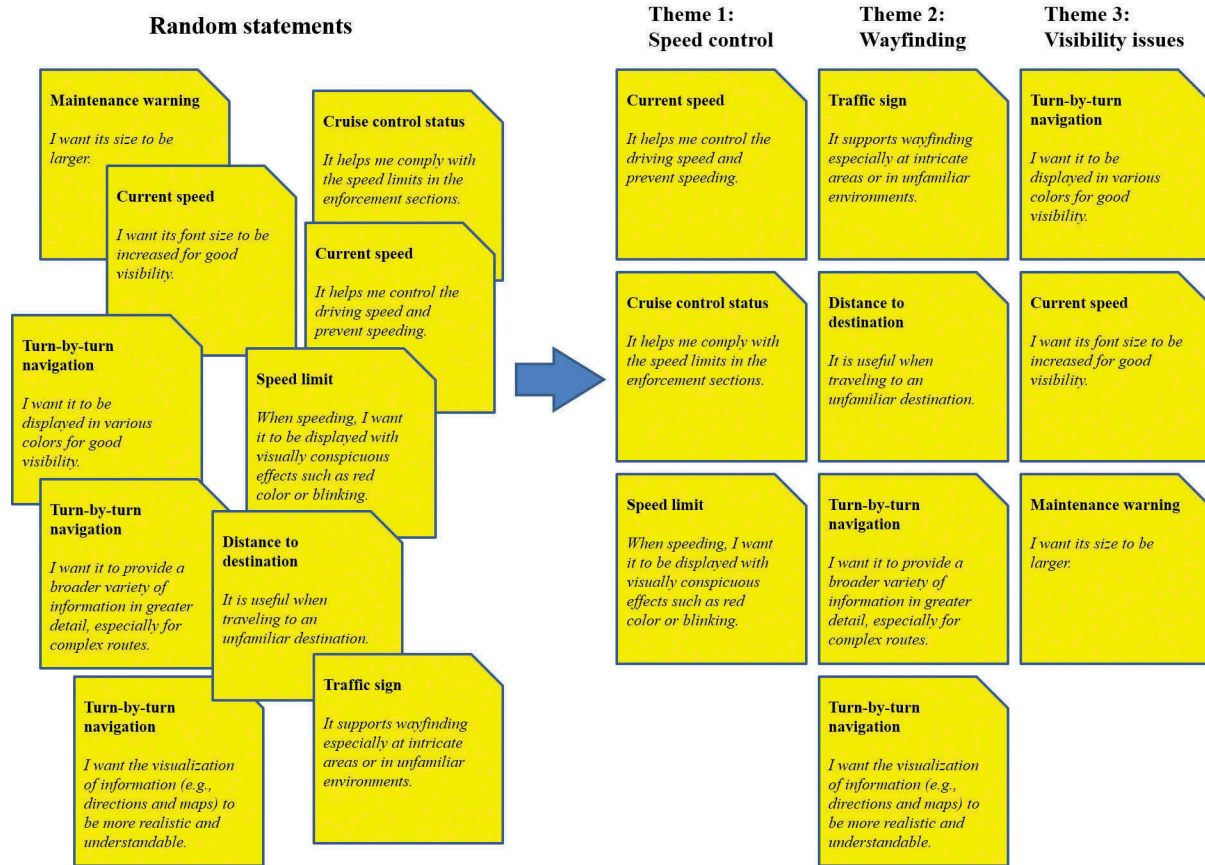


Figure 1. An illustration of the affinity diagram analysis.

identical to the methods of ‘contextual analysis’ described in Hartson and Pyla (2012).

For each of the themes/topics identified from the affinity diagram analysis, design implications were derived from the corresponding simple statements through deductive inferences. The deductive inference process involved a logical syllogism with each simple statement being the major premise and the existing HF/HCI design principles and guidelines being the minor premises – the deductive inference process was based on the ‘design requirements extraction’ method described in Hartson and Pyla (2012).

3. Results

Tables 2~12 summarize the participants’ responses to the survey regarding the context of use and design improvement

points for each of the eleven information items. Each table provides simple statements obtained by refining the participants’ original comments and their associated frequencies.

4. Discussion of the survey results

The current study conducted a user survey on the context of system use and user-perceived design improvement points for the existing commercial automotive HUD systems. A set of eleven high-priority HUD information items were considered. The survey participants were experienced automotive HUD users.

As a result of the affinity diagram analysis conducted on the participants’ statements dataset (Tables 2~ 12), a total of eleven themes/topics were identified. They were: 1) speed control, 2) highway driving, 3) engine/transmission control,

Table 2. Summary of participants’ survey responses: current speed (total 51 participants).

	Statements	Frequencies
Context of use	“I use it all the time while driving.”	24
	“It helps me comply with the speed limits in the enforcement sections.”	23
	“It is useful on the highway.”	5
Design improvement points	“It helps me control the driving speed and prevent speeding.”	4
	“When speeding, I want it to be displayed with visually conspicuous effects such as red color or blinking.”	5
	“I want its font size to be increased for good visibility.”	4
	“I want it to be displayed in various colors for good visibility.”	3
	“It slightly lags behind the actual value.”	3
	“I want it to be displayed on or near the driver’s normal line of sight or I want to be able to adjust its location.”	2
	“If displayed with speed limit, it would enhance driving safety.”	1
	“I want it to be displayed with RPM and gear position.”	1
		1

Table 3. Summary of participants' survey responses: speed limit (total 36 participants).

	Statements	Frequencies
Context of use	"It helps me comply with the speed limits in the enforcement sections."	16
	"It is useful on the highway."	4
Design improvement points	"It is useful when I overlook or have difficulty detecting the road sign due to the unfamiliarity with the driving route."	2
	"When speeding, I want it to be displayed with visually conspicuous effects such as red color or blinking."	7
	"I want the HUD system to provide accurate speed limit information through fast and automatic updates of the on-board navigation system."	6
	"I want it to be displayed all the time, regardless of whether the navigation system is activated or not."	4
	"I want it to be displayed only in the actual enforcement sections."	3
	"I want to be able to choose whether or not to display it."	2

Table 4. Summary of participants' survey responses: turn-by-turn navigation (total 48 participants).

	Statements	Frequencies
Context of use	"It supports wayfinding especially at intricate areas or in unfamiliar environments."	22
	"It keeps me informed of the route even when the center console display is occupied by information items other than navigation (e.g. audio player, digital multimedia broadcasting)."	1
Design improvement points	"I want it to provide a wider variety of information in greater detail, especially for complex routes (e.g. intersections, interchanges)."	13
	"I rarely use HUD navigation, since the provided information is inaccurate one transferred from the on-board navigation system of poor/unreliable wayfinding performance. Instead, I mostly use non-HUD navigation systems such as smartphone applications and aftermarket devices."	8
	"On-board navigation systems of foreign cars should be improved to provide accurate and efficient route guidance in the domestic road/traffic conditions."	6
	"I want it to be displayed in various colors for good visibility."	3
	"I want the visualization of information (e.g. directions and maps) to be more realistic and understandable."	3
	"I want the HUD system to provide accurate information reflecting frequently changing road traffic conditions through fast and automatic updates of the on-board navigation system."	2
	"I want it to show a brief mark of the current location."	2
	"Adding the cardinal points would be useful for grasping the driving direction."	1
	"It would be of great help to provide a lane change instruction as well."	1

Table 5. Summary of participants' survey responses: maintenance warning (total 26 participants).

	Statements	Frequencies
Context of use	"It allows immediate detection of the vehicle's problems and timely measures against them, which might have been overlooked otherwise."	5
Design improvement points	"I not only want a warning sign but also a detailed explanation for the problem and its solution."	6
	"I want its size to be larger."	1

*Note: Maintenance warning refers to the warning sign presented when maintenance is required (for tire pressure, engine oil, brake system, coolant, and so on).

Table 6. Summary of participants' survey responses: cruise control status (total 32 participants).

	Statements	Frequencies
Context of use	"It is useful on the highway."	16
Design improvement points	"It helps me comply with the speed limits in the enforcement sections."	3
	"It does not have to be displayed on the HUD, because it just clutters the limited space of the HUD screen."	3
	"When I adjust the cruise control settings, I want it to be immediately displayed without lags."	2
	"I want it to be displayed along with speed limit."	2

Table 7. Summary of participants' survey responses: audio player status (total 29 participants).

	Statements	Frequencies
Context of use	"It helps me check and manipulate (e.g. search, select, change) currently playing audio tracks or radio stations."	20
	"I frequently use it, as I always listen to music while driving."	7
Design improvement points	"It does not have to be presented on the HUD because it just obstructs my field of view."	1
	"I want the volume level to be displayed as well."	1
	"I want its font size to be increased for good visibility."	1

*Note: Audio player status refers to information about currently playing audio tracks or radio stations (e.g. title of a song, radio frequency).

Table 8. Summary of participants' survey responses: traffic sign (total 17 participants).

	Statements	Frequencies
Context of use	"It supports wayfinding especially at intricate areas or in unfamiliar environments."	5
	"It is useful on the highway."	4
	"It is useful when I overlook or have difficulty detecting the road sign due to the poor visibility of the road ahead."	2
Design improvement points	"I want the HUD system to provide accurate information reflecting frequently changing road traffic conditions through fast and automatic updates of the on-board navigation system."	3
	"I want to be able to turn it on or off regardless of whether the on-board navigation system is operating or not."	1
	"It should be designed to match up with the actual roadway signs."	1

*Note: Traffic signs include regulatory signs (e.g. no entry signs, no parking signs), warning signs (e.g. sharp curve, falling rocks, slippery road, bumpy road, fog area), and guide signs (e.g. street names, expressway signs).

Table 9. Summary of participants' survey responses: distance to destination (total 17 participants).

	Statements	Frequencies
Context of use	"It is useful when traveling to an unfamiliar destination."	5
	"It helps me estimate arrival times and schedule trips."	3
Design improvement points	"It does not have to be displayed on the HUD because I can easily get it from the on-board navigation or sometimes from the road signs."	3
	"I want the HUD system to provide accurate information reflecting frequently changing road traffic conditions through fast and automatic updates of the on-board navigation system."	2

Table 10. Summary of participants' survey responses: gear position (total 18 participants).

	Statements	Frequencies
Context of use	"It is useful for checking and shifting gears in manual shift mode."	7
	"It is useful for sporty driving."	5
	"It is useful for economical driving."	3
Design improvement points	"Since I usually drive in automatic mode, it is not that necessary."	5
	"I want it to be displayed with engine RPM."	3
	"I do not use it often."	2
	"I want to be able to choose whether or not to display it."	2
	"I want it to be displayed even when I drive in automatic mode (e.g. D1~ D7)."	2
	"I want its visual design to be aesthetically more appealing."	2

Table 11. Summary of participants' survey responses: RPM (total 12 participants).

	Statements	Frequencies
Context of use	"It is useful for sporty driving."	5
	"It is useful for checking and shifting gears in manual shift mode."	4
	"It is useful for economical driving."	1
Design improvement points	"I want it to be displayed only in sport mode, not in usual automatic mode."	2
	"It does not have to be displayed on the HUD because I can estimate the RPM by hearing the engine sound."	1
	"It slightly lags behind the actual value."	1
	"I want its design to be visually more outstanding for driving pleasure."	1
	"I want it to be displayed with gear position."	1

Table 12. Summary of participants' survey responses: low fuel warning (total 26 participants).

	Statements	Frequencies
Context of use	"It allows immediate identification of the low fuel level."	2
Design improvement points	"It does not have to be displayed on the HUD, because checking the instrument panel or hearing auditory warning is enough."	5
	"I want it to display the distance to empty all the time regardless of the fuel level."	1
	"I want it to display the instantaneous fuel consumption rate as well, which would be useful for economical driving."	1
	"It is currently displayed where the turn-by-turn navigation is being presented, so drivers may become distracted or confused by the warning, and may miss the navigation instructions."	1

4) wayfinding, 5) sign/warning detection, 6) audio player control, 7) accuracy issues of HUD information items, 8) individual- and context-specific HUD information needs, 9) visibility issues of HUD images, 10) visual aesthetics of HUD interfaces, and 11) HUD location and layout issues. Each theme/topic is discussed in what follows, focusing on frequent user responses and opportunities for design improvement and/or academic research.

4.1. Speed control

Many survey participants responded that checking current speed, speed limit and/or cruise control status through HUD helped control the vehicle speed, and, especially, comply with the speed limits (Tables 2, 3 and 6). Since speed control is a fundamental element of driving, which continuously requires the driver's attention, providing relevant information at an easily accessible location using HUD would be beneficial. Multiple previous studies (Doshi et al., 2009; Liu & Wen,

2004; Rutley, 1975; Sojourner & Antin, 1990) showed that drivers were better at controlling the vehicle speed and adhering to the speed limit when using HUD than HDD. Rutley (1975) stated that drivers' enhanced awareness of the speed-related information through HUD could be particularly useful considering the drivers' frequent misjudgment of the vehicle speed.

Some participants wanted current speed and speed limit to be displayed with conspicuous effects such as red color or blinking in case of speeding (Tables 2 and 3). Color- or blink-coding seems to be a good design strategy in that they are well suited for vehicle warnings requiring immediate attention (Green, Levison, Paelke, & Serafin, 1993; ISO/TR 16352, 2005; SAE J2831, 2012; Stevens et al., 2002), and, they can provide additional information without visual clutter. However, such visual effects in HUD should be carefully applied not to compel too much attention or annoy the drivers (Menozzi, Hofer, Näpfli, & Krueger, 2003; Tufano, 1997; Ward & Parkes, 1994). One possible design solution

may be to change the visual conspicuity (e.g. changing colors or blinking frequencies), depending on the vehicle speeds. Besides the user demands, given the importance of speed control in driving, further efforts may be warranted to develop new HUD concepts for supporting it. Indeed, some prototype HUD systems have been proposed so as to support longitudinal speed control. Examples include a HUD for presenting oncoming traffic volume (Charissis & Papanastasiou, 2010) and that for visualizing the vehicle braking distance (Plavšić et al., 2009).

4.2. Highway driving

Some participants commented that accessing current speed, speed limit, cruise control status and traffic sign through HUD was particularly useful on the highway (Tables 2, 3, 6 and 8). These results seem to reflect that 1) reducing information access costs for speed control-related information is important on the highways where speeding occurs frequently and speed cameras are mainly installed, and 2) rapidly detecting traffic signs is essential for correct and safe vehicle maneuvers at interchanges and ramps and in the areas with potential hazards. Also, since the driver's visual scanning area becomes narrower and viewing distance becomes farther at higher speeds (Park, Lee, Seo, & Choe, 2016; Sanders & McCormick, 1993), HUD would be particularly beneficial for highway driving.

Future study might explore new HUD functions and interface designs for highway driving. Especially, observational studies of drivers on highways would be useful for identifying drivers' information needs in that particular driving environment. Also, adaptively controlling the HUD focal distance and HUD image location according to the driving environment may improve the usefulness of automotive HUD. Relatedly, Park et al. (2016) have suggested that HUD images be focused at a farther distance and located closer to the center of line-of-sight at higher speeds to minimize eye transition/accommodation.

4.3. Engine/transmission control

Some study participants responded that checking gear position and RPM through HUD was useful for gear shifting in manual shift mode, and for sporty or economical driving (Tables 10 and 11). These responses are understandable as the driver needs to check gear position and RPM frequently in such driving contexts.

One possible design idea for automotive HUD is to change the HUD interface adaptively, according to the driving mode. Indeed, some performance vehicles (e.g. BMW M-series vehicles, Chevrolet Camaro and Corvette, and Mercedes-AMG vehicles) allow drivers to switch to a HUD interface displaying sporty driving-specific contents, such as shift light (indicating that the vehicle's best performance level has been reached to shift to higher gear), lateral acceleration (G-force), and lap and race timers. Similarly, HUD interfaces specific to other driving modes could be developed. For example, a HUD interface displaying instantaneous fuel consumption rate

(Table 12), or gear shift indicator for fuel efficiency would support economical driving.

4.4. Wayfinding

Many participants commented that HUD navigation information items were useful when finding their way especially in complex roads or new/unfamiliar places (Tables 4, 8 and 9). Wayfinding requires parallel visual information processing of navigation information and road scene ahead, and, thereby, demands considerable attentional resources and working memory processes (Montello & Sas, 2006). HUD seems to lighten such cognitive demands, by reducing information access costs for the parallel processing. Indeed, HUD was regarded as the best medium for displaying turn-by-turn navigation (Green, 1996), and HUD navigation showed lower reaction times, better awareness of surroundings, less navigational errors, and higher preference ratings than HDD navigation (Burnett, 2003; Jose, Lee, & Billingham, 2016; Srinivasan & Jovanis, 1997).

Some survey participants wanted detailed navigation instructions for complex routes (Table 4). Also, a few wanted a more realistic and understandable HUD interface design that utilizes visual elements resembling the real road environment and objects in it (Tables 4 and 8). Future research studies are needed to investigate how the level of details and that of realistic quality of the interface affect the understandability of a HUD-based navigation system and the relevant task performance. A detailed/realistic HUD interface for navigation may provide advantages over a simple/abstract one in terms of understandability; yet, it may cause side-effects, such as visual clutter and driver distraction (Lee et al., 2014).

4.5. Sign/warning detection

The survey results indicated that HUD facilitated timely detection of road sign (Tables 3 and 8) and vehicle maintenance warning (Tables 5 and 12). This reflects the main advantage of HUD over HDD, that is, the faster detection of visual information to better support situation awareness (Charissis & Papanastasiou, 2010; Liu & Wen, 2004; Sojourner & Antin, 1990; Yang et al., 2018). HUD would be a good medium for presenting time-critical information items in future vehicles, such as take-over request in automated driving (Cha & Park, 2006; Politis, Brewster, & Pollick, 2015; Walch, Lange, Baumann, & Weber, 2015).

As for maintenance warning, several participants responded that they wanted to obtain details through HUD, such as additional explanations and suggested solutions (Table 5). This user need seems to be in line with the existing design recommendation that in-vehicle warning symbols, despite the advantages of fast recognition and language independence, need to be supplemented with textual information or action instructions to be used effectively and prevent misunderstanding (Baber & Wankling, 1992; ISO/TR 16352, 2005). Despite the utility of additional detailed information, however, research is needed to develop display methods/interfaces that would safely and effectively deliver such additional information as presenting much information on HUD is likely

to be unsafe or undesirable in many driving situations (Charissis & Papanastasiou, 2010; Stevens et al., 2002). Also, individual drivers' different preferences need to be considered in determining the levels of detail for presenting the information (ISO 9355-1, 1999).

4.6. Audio player control

The survey results revealed that one of the main functions of commercial automotive HUD systems was to support audio player control (Table 7). This indicates that listening to music and other audio contents, although not pertaining to the primary driving tasks, is an important element of the overall driving experience (Beck & Park, 2018; Knobloch & Zillmann, 2002).

Future studies may explore new HUD interface designs for audio player control. In doing so, interface designs from other domains, for example, the 'Cover Flow' interface in desktop and mobile products for visually flipping through albums/tracks, could be tested for HUD implementation, and be compared with the existing scrolling list interface. HUD may also be used to present other information related to the audio player system, such as volume level (Table 7), progress bar, lyrics, or playing mode (e.g. shuffle, repeat). The necessities/utilities of such HUD features need to be examined.

4.7. Accuracy issues of HUD information items

The survey results revealed two information accuracy issues in the existing HUD systems. One is the inaccuracy of HUD navigation/traffic information, resulting from the delayed updates and poor wayfinding performance of the on-board navigation system (Tables 3, 4, 8 and 9). Since the existing Original Equipment Manufacturer (OEM) HUD systems display what has been already stored in the on-board navigation systems, these issues should be addressed by their regular/automatic updates and continual performance improvements. Alternatively, HUD systems could utilize information from navigation applications of mobile devices or actual roadway signs. A camera-based traffic sign recognition system would be needed to utilize roadway signs as a source of information (Grimm, 2013). Indeed, some OEM HUD systems (e.g. BMW, Jaguar, Land Rover, Mercedes-Benz, and Volvo) can display speed limit and no overtaking signs detected by such recognition system. Future HUD systems may be designed to combine/fuse information from multiple sources to improve the accuracy of the presented information.

The other issue is the time lag that exists in presenting current speed, cruise control status and RPM on the HUD in real-time (Tables 2, 6 and 11). Lagged systems are known to create considerable stress for the operator and degrade task performance due to the unavailability of feedback from earlier actions to help plan the current action and the increased cognitive demands required to anticipate the future state (Scholcover & Gillan, 2018; Wickens, Hollands, Banbury, & Parasuraman, 2013). Considering that the commented information items are directly relevant to the primary driving tasks, the latency should be reduced not to adversely affect task performance, nor to make drivers look down to HDD instead.

4.8. Individual- and context-specific HUD information needs

The survey results overall indicated that information needs of HUD users varied significantly from one user to another, and, also from one context to another (Tables 3, 10, 11 and 12). Some of the survey participants responded that they wanted to be able to decide what to display with their HUD systems and also specify the conditions for turning on/off certain HUD information items (Tables 3, 8, 10 and 11). Some information items were even considered unnecessary because of information redundancies (easily obtainable from HDD or auditory signal), individual driving styles, and concerns about clutters and/or obscurations (Tables 6, 7, 9, 10, 11 and 12).

One way to address the individual- and context-specific nature of the user information needs is to provide customizable features to HUD systems. While some existing HUD systems can be customized, users are only allowed to turn on/off only a certain part of the entire information provided or switch between several sets of information items predetermined by the manufacturer (e.g. GM [Buick, Cadillac, Chevrolet, GMC], Honda, Mercedes-Benz, Toyota) – they are not allowed to turn on/off each information item one by one. Future HUD systems may need to be more flexible to better address the individual- and context-specific HUD information needs.

Despite the likely benefits of customizable HUD systems, however, basic research needs to be conducted to ensure safety and utility of such systems. Some research questions are as follows: 1) User-customization of a HUD system should be restricted within certain safety limits so as not to compromise driving safety (Stevens et al., 2002; Yayici, 2014). How can the safety limits on the amount of information displayed by an automotive HUD system be determined for each individual driver? Can drivers accurately determine the safety limits for themselves? 2) Can users determine HUD information items they need and/or specify the display-on/off conditions? In other words, do they know what they need/want and when? If not, how can a HUD system discover/learn the user's information needs/wants? 3) How can a HUD system be customized in an easy and efficient manner? What is the user interface that best assists the drivers in specifying their personal settings?

4.9. Visibility issues of HUD images

Some participants suggested using various colors or large-sized images for improving the visibility of HUD images (Tables 2, 4, 5 and 7). Visibility is a critical human factors design issue in developing automotive HUDs, and, previous studies have provided relevant design suggestions (Ahlgren & Kudrick, 2007; Choi, Lee, Lee, & Park, 2013; Inuzuka, Osumi, & Shinkai, 1991; Park et al., 2016).

Some research questions concerning the visibility of HUD images include the following: 1) Visibility of HUD images would deteriorate with visual complexity of the background, especially with dynamic interference from the outside road environment (Stevens et al., 2002; Ward & Parkes, 1994). How can HUD images be adaptively displayed to be visible

in all of the potential viewing conditions? 2) Large-sized HUD images can add to visibility, but at the same time, compromise compactness of a display (ISO/TR 16352, 2005). How can a proper balance for their size be achieved? 3) Visibility issues would require careful consideration of individual differences. Some OEM HUD systems (e.g. Genesis, Hyundai, KIA) allow users to choose from three font colors (white, orange and green) and three font sizes (small, medium and large) for displaying current speed in order to accommodate the needs/preferences of different drivers. How much adjustability is required to achieve a high level of population accommodation in displaying each major information item?

4.10. Visual aesthetics of HUD interfaces

A few participants wanted more appealing visual aesthetics for automotive HUD interfaces (Tables 10 and 11). Appealing visual aesthetics is in general known to positively affect usability, user performance, or overall user experience (Moshagen, Musch, & Göritz, 2009; Reppa & McDougall, 2015; Sonderegger & Sauer, 2010; Tractinsky, Katz, & Ikar, 2000). Aesthetic preference varies greatly across different individuals – age, gender, personality, cultural background and experiences are known to affect the responses to aesthetics (Crilly, Moultrie, & Clarkson, 2004; Sonderegger & Sauer, 2010). Research studies are needed to understand the aesthetic preferences of the automotive HUD user population.

4.11. HUD location and layout issues

A few participants commented that they wanted current speed to be displayed on the driver's normal line-of-sight (Table 2). This comment seems reasonable because displaying the high-priority information close to the sightline is generally recommended (Bhise, 2011; Green et al., 1993; ISO/TR 16352, 2005; Stevens et al., 2002). However, it has also been recommended that HUD image should be slightly away from the direct line-of-sight to avoid cognitive capture or visual obscuration (Bergman, 2011; Horrey & Wickens, 2004; Sojourner & Antin, 1990; Stevens et al., 2002; Ward & Parkes, 1994; Watanabe et al., 1999; Wittmann et al., 2006). The trade-offs between these two design considerations should be carefully studied in future research efforts.

Additionally, there was an issue of multiple information items being displayed overlapped with one another (Table 12), which needs to be resolved to avoid potential problems, such as driver distraction and/or information loss. If multiple information items are to be presented simultaneously, their relative priority may be considered to optimize their layout – for example, by emphasizing the higher-priority items to be more visually accessible, without being overlapped/blocked by others (ISO/TR 16352, 2005; SAE J2831, 2012). Alternatively, allowing drivers to deactivate any or all of the items presented on HUD on an ad hoc basis (SAE J2831, 2012) could, to some extent, address the issues of information items being overlapped.

Lastly, many commercial HUD systems currently allow users to adjust the location of the displayed items as a whole, but not individually. An additional adjustability feature for controlling the location of each displayed item

may help better accommodate drivers with different preferences (Guo et al., 2014).

5. Concluding remarks

This study reported the results of a user survey on the usage contexts and design improvement points of the existing automotive HUD systems. A set of eleven high-priority HUD information items displayed by the existing commercial HUD systems were considered. Drivers with significant prior HUD use experience participated in the user survey. The major study findings of the current study and their design implications are summarized in what follows:

- Speed control – many participants responded that checking current speed, speed limit and/or cruise control status through HUD helped control the vehicle speed and comply with the speed limits; some participants wanted conspicuous effects such as red color or blinking in case of speeding; changing the conspicuity of such visual effects depending on the vehicle speeds may support speed control without drawing too much attention or annoying the drivers.
- Highway driving – some participants commented that accessing current speed, speed limit, cruise control status and traffic sign through HUD was useful on the highway; observational studies of drivers on highways may help identify drivers' information needs in that particular driving environment; also, controlling the HUD focal distance and HUD image location according to the driving environment may help reduce eye transition/accommodation.
- Engine/transmission control – some participants responded that checking gear position and RPM through HUD was useful for gear shifting in manual shift mode, and for sporty or economical driving; changing the HUD interface adaptively, according to the driving mode may help support the mode-specific engine/transmission control.
- Wayfinding – many participants commented that HUD navigation information items were useful for wayfinding, especially in complex roads or new/unfamiliar places; some participants wanted detailed navigation instructions for complex routes, and a few wanted a more realistic/understandable HUD interface design; future studies are needed to develop a detailed/realistic HUD interface that can support wayfinding without causing visual clutter and driver distraction.
- Sign/warning detection – the survey results indicated that HUD facilitated timely detection of road sign and vehicle maintenance warning; for maintenance warning, several participants wanted to obtain details through HUD, such as explanations/solutions; research is needed to develop display methods/interfaces to safely and effectively deliver such additional information, with consideration of individual drivers' different preferences for the levels of detail.
- Audio player control – the survey results revealed that one of the main functions of automotive HUD systems

was to support audio player control; interface designs and/or features of audio player systems in other domains could be tested for HUD implementation in terms of their necessities/utilities.

- Accuracy issues of HUD information items – the inaccuracy of HUD navigation/traffic information could be addressed by regular/automatic updates and improvements of the on-board navigation systems; alternatively, HUD systems could utilize information from multiple sources, such as mobile devices or roadway signs to improve the accuracy; the time lag in presenting current speed, cruise control status and RPM on HUD should be addressed not to adversely affect driving performance, nor to make drivers use HDD instead.
- Individual- and context-specific HUD information needs – information needs of HUD users varied significantly from one user to another, and, also from one context to another; allowing users to decide what to display and to specify the display-on/off conditions for each of the entire information items may help address the individual- and context-specific nature of the user information needs.
- Visibility issues of HUD images – some participants wanted various colors or large-sized images for good visibility of HUD images; future studies concerning the visibility issues need to consider 1) all of the potential viewing conditions, 2) trade-off between visibility and visual clutter, and 3) individual differences.
- Visual aesthetics of HUD interfaces – a few participants wanted more appealing visual aesthetics for HUD interfaces; research studies are needed to understand aesthetic preferences of the automotive HUD user population.
- HUD location and layout issues – a few participants wanted current speed to be displayed on driver's normal line-of-sight; also, there was an issue of multiple information items being displayed overlapped with one another; allowing users to adjust the location of each of the displayed items individually may help accommodate drivers with different preferences for the HUD location/layout.

The above study findings and design implications would help improve the existing HUD systems in their functions and their interface/interaction design, and, thereby, contribute to enhancing the overall user experience of automotive HUD users.

Finally, a couple of ideas for further research on user experience of automotive HUD systems are provided here: first, the current study results pertain to the existing commercial automotive HUD systems; as the automotive HUD systems evolve along with the advanced driver-assistance systems (ADAS) and in-vehicle information systems (IVIS), and, also the autonomous vehicle technology, the contexts of system use and the users' wants and needs will change accordingly. Thus, continuous follow-up research will be necessary to update the current study results.

Second, the current study utilized the written survey method for data collection. While written surveys have been

used to understand product use practices and user wants and needs in a variety of product design contexts (Goodman et al., 2012; Rubin & Chisnell, 2008), the UX literature generally emphasizes the importance of on-site observations of users and product use practices in natural settings (Goodman et al., 2012; Hartson & Pyla, 2012). Additional studies based on on-site observations may provide further design insights.

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No potential conflict of interest was reported by the authors.

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